
15. Demystifying green bonds

*Kathrin Berensmann, Florence Dafe and
Nannette Lindenberg*

1. INTRODUCTION

Governments, investors and the media have hailed green bonds – debt instruments intended to finance environmentally friendly projects – as a key instrument of climate finance because they could efficiently tap the trillions of dollars held in global bonds markets and by institutional investors. Representatives from development banks (DBs), for instance, often portray green bonds as ‘essential to confronting climate change’ (World Bank, 2015b) while institutional investors emphasize that green bonds make it possible to invest for sustainability and also achieve the financial performance that shareholders and policyholders expect (Zurich Insurance Group, 2014: 26).

A quick glance at the data does indeed indicate that global bonds markets are ‘greening’. Two categories of green bonds have emerged in bond markets, namely green bonds which are labelled, that is certified as green, and unlabelled green bonds, which are not.¹ A sub-category are climate bonds, where the proceeds are linked to projects that address climate change. Annual labelled green bonds issuance has risen steadily – from US\$ 2.6 billion in 2012 to US\$ 160 billion in 2017 (CBI, 2015a, 2017a, 2018a; 2018b). The Climate Bonds Initiative (CBI), a non-governmental organization (NGO) that promotes green bonds markets, estimates that in 2018 issuance of labelled green bonds will reach USD 250 billion (CBI, 2018b).

In this chapter we probe the potential of green bonds to contribute to sustainable development by financing investments that support the environment, notably through combating climate change and its impact. The extant literature on green bonds is limited. The vast majority of the literature has been published by three types of actors: issuers of green bonds; investors; and NGOs. There is hardly any academic research on the topic.² Moreover, while publications on green bonds are increasing, the authors of studies are barely addressing one another’s work and often focus on one dimension of green markets development such as market size, while neglecting others.

This chapter contributes to the emerging literature on green bonds by connecting different strands of the literature and by providing a more holistic assessment of the potential contribution of green bonds to sustainable development. In assessing the potential contribution of green bonds to sustainable development, we examine the size of the green bonds market and the factors that have driven its development as well as those that impede further development. Particular emphasis is placed on delineating the governance structures of the green bonds market – which have been heavily criticized by institutional investors and other financial market experts – and their consequences. We also use data from a variety of secondary sources and semi-structured interviews with issuers, investors and scholars of climate finance to examine the claim that the green bonds market can significantly contribute to sustainable development.

Our conclusions only partly support this claim. We find that there is significant potential for growth in the green bonds market as suggested by the market's increase in size and geographic reach and the broadening of the investor base. However, there are also limitations such as shallow secondary markets, a narrow investor base and 'greenwashing', the risk of green bonds' proceeds not being used to finance genuinely green projects. A major reason for these limitations is the weakness of the framework that governs green bonds markets. In particular, there are shortcomings with respect to the rules for standardization and disclosure: the governance structures for green bonds have limited transparency and accountability. In order for the green bonds market to mature with integrity, these shortcomings must be addressed.

The chapter proceeds as follows. In section 2 we first briefly describe the major types of green bonds and issuers and then discuss the significant growth of the green bonds market. In section 3 we explain the major impediments to their further development and argue that addressing weaknesses in the framework for governing green bonds is essential for the development of green bonds markets. In the final section, we present policy implications with a particular focus on measures for improving the governance of the market for green bonds.

2. CAPITAL MARKETS, GREEN BONDS AND THE TRANSFORMATION TOWARDS A GREEN ECONOMY

There is broad agreement among scholars and policymakers that transitioning towards a low-carbon society requires significant investment of economic resources in ecologically sustainable development. It is

estimated that in order to limit global warming to 2°C, annual investments in clean energy must be quadrupled to reach US\$ 1 trillion each year until 2030 (Fulton and Capalino, 2014). At the same time, investments must be reduced in carbon-intensive sectors that lock in greenhouse gas emissions for decades. In 2015, leaders from 195 countries and the European Union (EU) negotiated a climate agreement to limit global warming to at least 2°C (UNFCCC, 2015). In order to channel savings into sectors and projects that support economically sustainable development it will be important to intensify the use of adequate financing instruments (Berensmann and Lindenberg, 2018). One such instrument – the green bond – has been hailed as key to green financial markets for increasing the share of green investments.

2.1 What Are Green Bonds?

Green bonds have no universal definition: they are self-labelled and have no legal parameters. In 2014, a group of major green bonds issuers, underwriters and investors developed the Green Bond Principles (GBPs) – voluntary process guidelines intended to promote integrity in the green bonds market. The GBPs broadly define green bonds as any type of bond instrument whose revenues are exclusively used to partly or completely finance or re-finance new and/or existing ‘eligible’ green projects.

To delineate developments in the global market for green bonds, it is helpful to consider five different characteristics: (1) the type of green bond; (2) the size of the market and targeted areas; (3) the issuer; (4) the objective of the issuance; and (5) the market regulations (ICMA, 2017).

2.2 Types of Green Bonds

Green bonds markets trade four major types of bonds, which mainly differ in the way the assets that securitize the bond are grouped (ICMA, 2017). The majority are green ‘use-of-proceeds’ bonds, which means that the proceeds of this type of bond are earmarked for green projects in the issuer’s portfolio, and the bond’s recourse is to the issuer’s entire balance sheet. When a green ‘use-of-proceeds’ bonds is issued, net revenues from issuances are moved to a sub-portfolio of the issuer’s general account or otherwise tracked. The net proceeds from these bonds are then disbursed to any eligible green projects. Since use-of-proceeds bonds are backed by the issuer’s complete balance sheet, credit-rating agencies do not rate individual bonds: green bonds of this type receive the same credit rating as their issuer.³

The other types of green bonds are also generally earmarked for green

projects. However, the recourse of the second type, the ‘use-of-proceeds revenue’ bond, is limited to the issuer’s pledged revenue streams. The third type is the ‘green project’ bond, which is used for specific green projects whose recourse is restricted to the project’s assets and balance sheet. The fourth type is the ‘green securitized’ bond, whose revenues are used for a group of green projects with recourse to a certain group of assets on the issuer’s balance sheet.

It is important to note that green bonds may be issued as ‘covered’ and ‘uncovered’ bonds. Covered bonds are bonds backed by assets on the issuer’s balance sheet; most investors of a covered bond have preferential claims in the event of default. Although most green bonds have been issued as uncovered bonds, covered bonds have several advantages, one of which is their reduced investment risk. Since covered bonds tend to be highly regulated, they usually have better credit ratings than uncovered bonds and lower funding costs. Moreover, covered bonds are subject to stricter legislation and their assets must be transparent – guaranteeing the quality of their collaterals. Green bonds could be used as covered bonds if public sector guarantees were added to existing legislation (Damerow et al., 2012; Kreditanstalt für Wiederaufbau, 2016). A rare example of a covered green bond is that issued by the Berlin Hyp in 2015 (Berlin Hyp, 2016).

2.3 The Size of the Market and the Targeted Areas

Developments in the green bonds market are also revealed by the size and areas financed through bonds issuance. According to the GBP, to assure that the financed projects actually provide sustainable environmental benefits, projects to be financed by green bonds must fall into one of the following broad categories: renewable energy, energy efficiency, sustainable waste management, sustainable land use, biodiversity conservation, clean transportation, sustainable water management or climate-change adaptation (ICMA, 2017).

The CBI is the most-cited source of estimations related to the size of the green bonds market. According to the CBI US\$ 82 billion of labelled green bonds were issued in 2016 (CBI, 2018b). In the same year, outstanding labelled and unlabelled green bonds which CBI calls climate aligned bonds amounted to US\$ 895 billion, 221 billion of which were labelled green bonds. In 2012, less than 5 per cent of climate aligned bonds were labelled green bonds. These data show that the green label has significant market potential (CBI, 2017a).

In 2016, the proceeds of labelled green bonds were mainly utilized for renewable energy and green buildings projects (64 per cent of total). The use of proceeds for transport amounted to 14 per cent and for water 10 per

cent. Land use, waste management and adaptation each made up only 3–5 per cent of total use of proceeds in 2016 (CBI, 2017a: 18).

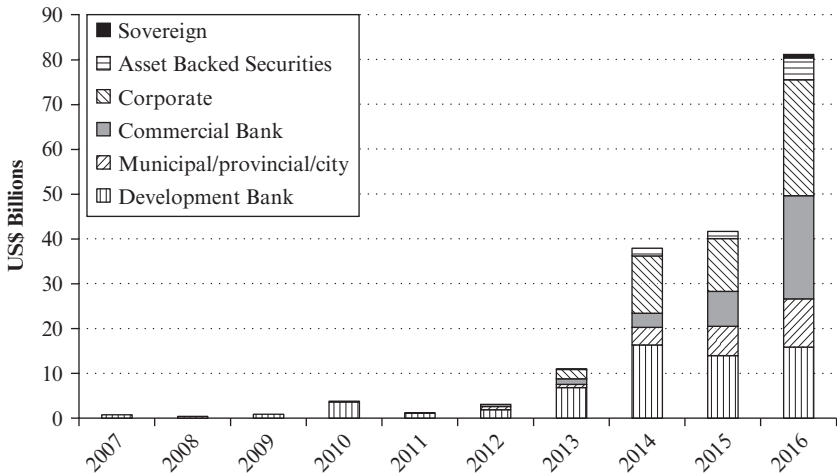
While these numbers are impressive, the green bonds market remains tiny in relation to the multi-trillion-dollar corporate bonds market.⁴ Green bonds make up less than 1 per cent of global bonds (G20 GFSG, 2016: 3). Moreover, a lot hinges on the definition of what constitutes a ‘green’ or ‘climate-aligned’ project or sector. The ‘greenness’ of the targeted field or sector is often questioned, which makes it more difficult to objectively estimate the market’s size. For instance, any estimations of the market volume based on the assumption that rail transportation cannot be labelled ‘green’ – regardless of the energy source and where the rails might be laid – would yield much smaller numbers for the market’s size. The inclusion of nuclear power or even large hydropower projects in enumerating eligible ‘green’ projects is also contested.

Moreover, the bonds were partly issued in different currencies in 2016: labelled green bonds were mainly issued in US\$ (36 per cent) and Euros (38 per cent), while climate aligned bonds including both unlabelled and labelled bonds were quite often also issued in emerging market currencies such as the Chinese currency (32 per cent). The US dollar and the Euro accounted for only 26 per cent and 20 per cent of total, respectively (CBI, 2017a: 5 and 18). The greater diversity of currencies in the unlabelled green bonds market can be explained by the fact that in emerging countries many issuers do not label their bonds ‘green’.

2.4 The Issuers

The third aspect that describes the universe of green bonds are their sources. As Figure 15.1 shows, the landscape of issuers has become much more diversified in the last two years. After the European Investment Bank (EIB) issued the first green bonds in 2007, the number of labelled green bonds issuances grew slowly in the wake of the 2008 global financial crisis, when there was little appetite for new financial instruments. At first, DBs were the major players in the green bonds market. However, the share of green bond issuance from DBs in terms of total labelled green bond issuance decreased to about 21 per cent in 2017. Banks and municipalities began to enter the market in 2012, with corporations following a year later. Similarly, sub-sovereign issuance from Local Government and Government-backed Entities amounted to 21 per cent of total labelled green bond issuance in 2017. The increase of issuances by sovereigns was remarkable in 2017: while their issuance amounted to US\$ 780 million in 2016, they went up to US\$ 10.8 billion in 2017 (CBI, 2018a; 2018b).

Hence, the green bonds offer was much more complex in 2016 and 2017



Source: Data provided by CBI.

Figure 15.1 Annual labelled green bonds issuance by issuer type

than it was in the pioneering years. This complexity is not only the result of the increase in volume and market participants; it results above all from the new types of issuers.

As a matter of fact, more and more professional investors are becoming convinced that sustainability considerations can help to improve their risk management tools and help them to make better decisions about investments.⁵ The fear of assets being stranded due to the growing political, economic and environmental risks related to fossil fuel investments makes green investments more attractive. Unforeseen events like the Volkswagen emissions scandal in the automotive sector that came to light in 2015 and the risks of mass legal proceedings or loss of reputation are other strong drivers of this new dynamism.

2.5 The Objective of the Issuance

A final consideration for illustrating developments in the green bonds market and its potential is the underlying motivation for issuing green bonds. NGOs expect that green bonds will provide additional climate finance, while companies want to show that they are ecologically conscious by conducting 'green image campaigns'. Other important reasons to issue green bonds are to refinance existing projects and diversify the investor base.⁶

The diversification of the investor base is particularly relevant for

DBs against the backdrop of the bond-purchasing programmes of many central banks in the aftermath of the global financial crisis, which led to the dominance of central banks among potential buyers and a kind of 'monoculture' in the DBs' bonds market. A senior official from the KfW development bank explains that DBs view diversification of the investor base through green bonds issuances as a way of insuring against the time when central banks reduce their bond purchases.⁷ They prioritize establishing good relationships with alternative purchasers to safeguard against a future lack of investor demand.

The experience of the African Development Bank (AfDB) suggests that the issuance of green bonds may indeed help to diversify the investor base by attracting new, socially responsible investors (Duru and Nyong, 2016). Usually central banks and other official bodies buy about 75 per cent of the AfDB's benchmark bonds, but in October 2013 socially responsible investors bought 84 per cent of its US\$ 500 million green bonds issuance. Central banks and official institutions constituted only 28 per cent of the buyers: 43 per cent were asset managers, 28 per cent insurance companies and pension funds, and 1 per cent retail and private banks.

An additional motivation for DBs is raising awareness – altruistically, for the sake of climate-related investments, particularly in emerging and developing economies and to market their own climate-related activities. Furthermore, DBs wish and sometimes are mandated to build and scale up the market for green bonds, as in the case of the KfW.

However, it is not clear that mobilizing additional climate finance is the objective of green bonds issuers. Like traditional bonds, green bonds often serve to re-finance existing projects. We do not yet have enough evidence to suggest that green bonds attract more new financing than traditional bonds (Clapp et al., 2016: 8). In this sense, the terminology of the EIB, which in 2007 issued the first green bond called a 'climate awareness bond' (CAB) and continues to use this name, is actually more appropriate for DB's green bonds.

Nevertheless, the CBI argues that green bonds' true additionality is offering project developers simple, reasonably priced possibilities for refinancing which incentivize them to make an initial investment (Sonerud et al., 2015). According to the CBI, the leveraging effect of green bonds may be indirect rather than direct. The certainty of having good possibilities for offloading an initial investment after some years through the green bonds market eliminates the long holding period of the majority of green investments, which can deter investment.

2.6 Rules Governing the Green Bonds Market

A plethora of agents is involved in developing rules, especially regarding standards, for the green bonds market. Among the key actors are states and public and private issuers such as development and commercial banks, NGOs, research institutes and rating agencies. However, most of the impetus to formulate rules and standards has come from outside the public domain in the form of industry self-regulation.

The first standards designed for the green bonds market were the GBPs – voluntary guidelines for issuing green bonds, which were first formulated in 2014 and have been updated annually since then. The GBPs specify a number of categories for projects eligible for green bonds financing and recommend transparency and disclosure with respect to the use of the proceeds (ICMA, 2017). Issuers, investors and intermediaries in the green bonds market, as well as the International Capital Market Association (ICMA), formulate the GBPs. The guiding principle is industry self-regulation, with membership for the GBPs being restricted to organizations that have issued, underwritten (placed) or invested in a green bond. Others may apply for observer status regarding the formulation of guidelines but not be involved in the drafting process.

The GBPs are not the only effort to develop standards for the green bonds market. The CBI has also developed the ‘Climate Bond International Standards and Certification Scheme’, which includes a certification process, pre- and post-issuance requirements and a suite of sector-specific eligibility and guidance documents (CBI, 2017a). Working groups of technical and industry experts, supervised by an advisory board of institutional investors and environmental NGOs, develop the Climate Bond Standards. Another type of standard setter is the second opinion provider, often a research institute or consulting agency. Before a green bond is issued, its issuers can seek second opinions, independent assessments of their green credentials and the green bonds issuance framework. Credit-rating agencies that have developed green bonds indices have also promoted standards for green bonds by ensuring that only green bonds that fulfil certain criteria are included. Examples are the S&P Dow Jones, Solactive, Barclays MSCI and Bank of America Merrill Lynch (S&P, 2017).

At the end of 2017, the Association of Southeast Asian Nations (ASEAN) Capital Markets Forum (ACMF), including capital market regulators from ASEAN countries, implemented the ASEAN Green Bond Standards that are based on the GBP (ACMF, 2017).

The public sector has also attempted to formulate standards. Several DBs have not only sought to shape the GBPs by becoming members, but

have also led efforts regarding post-issuance reporting. The AfDB, the EIB, the International Finance Corporation (IFC) and the World Bank (IBRD) have been working to create a harmonized framework for reporting the impacts of projects that have been allocated green bonds proceeds (AfDB et al., 2015; Reichelt and Davies, 2015; IDFC and MDB, 2015). These DBs recommend that core indicators be used to report their impacts on energy efficiency and renewable energy projects.

The People's Bank of China Green Finance Committee has published the Green Projects Catalogue, which establishes rules for evaluating projects to ensure their eligibility for green bonds financing (Green Finance Committee of the China Society of Finance and Banking, 2015). In November 2017, China's Green Finance Committee (CGFC) and the European Investment Bank (EIB) published a white paper with the aim of increasing the comparability of green bond standards in China and the EU that is based on the Chinese Green Bond Endorsed Projects Catalogue, the project eligibility criteria of EIB's climate Awareness Bonds and the MDB-International Development Finance Club's (IDFC) Common Principles for Climate Mitigation Finance Tracking (EIB and CGFC, 2017; IDFC and MDBs, 2015).

Another example of a public attempt to govern the green bonds market are the guidelines published by the Securities and Exchange Board of India (SEBI, 2016). These guidelines include rules about third-party verification of green bonds by independent auditors. Other states and the EU are also pondering whether they should regulate the green bonds market, since it is not clear that industry self-regulation suffices to ensure its quantitative and qualitative growth (Kretschmer, 2016).

Overall, however, the formulation of standards has been driven far more by markets than by states. Multilateral bodies like the United Nations Environment Programme (UNEP) or the G20 also have not taken the lead in efforts to improve the green bonds market's governance – which may reflect the broader governance vacuum with regard to steering the green economy (Tienhaara, 2016). States' hesitance to more actively govern green bonds markets may reflect broader concerns about stifling the market's growth through standards and regulations (Kretschmer, 2016).

3. CHALLENGES FOR DEVELOPING THE GREEN BONDS MARKET

While both quantitative and qualitative criteria indicate that significant progress has been made in developing the green bonds market, we should be wary of painting too rosy a picture. The literature on green

bonds highlights a variety of obstacles to the sustainable growth of green bonds markets, notably, the weakly developed pipeline for green projects largely due to the absence of policies and regulations that create demand for green investments; the costs associated with labelling a bond 'green' – administrative certification, verification and monitoring; green investors' unwillingness to pay a 'green' premium and accept smaller returns for those bonds than for their traditional 'grey' alternatives; and in the case of emerging and developing economies, weakly developed capital markets and low credit ratings for potential green bonds issuers.⁸

The main challenge to the market's further development relates to the governance of the green bond market. While the academic literature examining the governance of the green bonds market is limited, there is a broad body of work by policymakers, market analysts, issuers and investors suggesting that the limited standardization and disclosure rules for green bonds are problematic.⁹ Although the market's governance framework is developing dynamically, it still has deficiencies. It is not clear that voluntary guidelines for transparency and disclosure can ensure that the market will mature with integrity. What is more, there is still no universal agreement among market participants, such as issuers and investors, about what constitutes a green bond and the characteristics required for projects to be eligible for green bonds financing.

One core problem has been missing harmonised standards. While several international and national taxonomies have been established, including the GBPs and the Climate Bonds Standard, the architecture of green bond standards at the international and national level is fragmented. Similarly, a number of voluntary standards and various instruments for certifying green bonds has been implemented, including second opinions, green ratings and green bond indices (Berensmann, 2017).

Finally, the continued lack of transparency and disclosure makes it difficult for regulators, NGOs and research institutions to assess market developments (Schneeweiß, 2016). Similarly, there are no legally binding standards for disclosing how green bonds proceeds are used, reporting the impact of projects funded by green bonds proceeds or independently assuring the green credentials. Monitoring is often recklessly neglected.

How might limited rules regarding standardization and disclosure affect the development of the green bonds market? One could argue that more governance and standardization would slow the market's dynamic growth. However, there is some risk that the uncontrolled increase of green bonds without stringent standards creates a market bubble and that the insecurity due to this development may eventually cause it to burst.

Moreover, closer examination of the effects of limited governance and standardization reveals that potential green bonds issuers might be hesi-

tant to issue a green bond due to the risk of having their green credentials challenged (KPMG, 2015). Universally accepted standards for projects financed by green bonds would reduce the likelihood that a particular bond's green credentials would be scrutinized. Enhanced standardization could also help to increase the pipeline of projects eligible for green bonds by providing potential issuers with clearer guidelines to design and select projects that could be financed through green bonds. Increased standardization and cheaper certification seem to be key to increasing the supply of green bonds.

In much the same way, enhanced standardization and disclosure rules might also be needed to broaden the investor base (Lindenberg, 2014). Investors may be hesitant to invest in green bonds because weak standards increase the possibility of challenges to a bond's green integrity and, by extension, its reputation (OECD, 2015). This is particularly true for institutional investors who have both huge amounts of money under management and strict investment criteria. In fact, in the Paris Green Bonds Statement, investors controlling US\$ 11.2 trillion encouraged experts to support the green bonds market by creating clear and independent standards, as well as introducing transparency rules about the use of their proceeds and their impacts (CBI, 2015b).

Broadening the investors base, which is currently dominated by the 'buy-and-hold' types, would also help deepen the secondary market for trading green bonds. A deeper secondary market in turn would make the green bonds market more attractive to investors.

Another way that weak rules governing green bonds can inhibit investment in the green bonds market is by limiting the scope to legally enforce green integrity (OECD, 2015). Moreover, the green bonds market's lack of clear standards may hinder investments in green bonds because investors have to make special efforts to examine the green credentials of a bond and its issuer, and an extra due diligence may be difficult to fulfil. Many investors have limited capacities for analysing green investments (OECD, 2015).

Enhanced and more consistent standardization and rules regarding disclosure are not just important for the green bonds market to quantitatively develop by increasing the scale and depth of the market; standards and rules for disclosure also help to qualitatively develop the green bonds market by reducing the risk of greenwashing, thereby enhancing market integrity. For instance, highly polluting companies must be prevented issuing green bonds to finance green projects in order to free up funds to subsidize grey operations.

The projects currently financed by green bonds proceeds exhibit different shades of green, ranging from pale green projects, with very limited

environmental impact, to dark green projects that significantly impact the environment. Weak standards and disclosure rules make it difficult to assess the environmental benefits of a particular bond. For instance, in 2014 a Massachusetts State agency sold green bonds and planned to use some of the proceeds to build a 725-space parking garage – one example of the risk of using green bonds proceeds for projects with dubious environmental benefits (Cherney, 2015). Another controversial case was a green bond sold by the French power company GDF Suez in 2014 to build a hydropower plant in Brazil that is deemed to have significant environmental and social costs, such as contributing to floods displacing tens of thousands of people and potentially significant greenhouse gas emissions (Banktrack, 2015; Cherney, 2015).

More comprehensive and binding governance is essential for the sustainable growth of green bonds markets. Moreover, with the increasingly global issuance of green bonds, international coordination to develop standards and disclosure rules are needed to avoid a ‘race to the bottom’ through regulatory arbitrage. It is also important to ensure that developing and emerging economies maintain some flexibility to adapt global standards to their local contexts, capacities and necessities (Berensmann, 2017). However, better global standards are still needed. Although improved financial governance through enhanced standardization and disclosure rules will not solve all the challenges in developing the green bonds market, without better governance it is unlikely to mature with integrity.

While better governance through enhanced and more consistent standardization and disclosure rules is essential for the green bonds market to develop, it will not solve all the challenges, such as that of pricing. At least in the primary market, green bonds are usually priced like their grey alternatives because investors are unwilling to assume a reduction of the bond’s return for the sake of making a green investment, despite the higher costs of issuance caused by the costs of green labelling.¹⁰ As a consequence, some companies with green credentials and projects have refused to issue green bonds because of the extra costs – and some highly polluting companies have issued green bonds (Jackson, 2016).

A study by Barclays (2015) suggests that enhanced demand for green bonds might allow green bonds to be more highly priced. Barclays found that investors are currently paying a premium to acquire green bonds in the secondary market, which is partly attributed to opportunistic pricing based on strong demand from environmental funds. Policies like improved standards and regulations to stimulate interest in green investments and raise the demand for green bonds may also help to address the pricing challenge.

One challenge for the development of the green bonds market that cannot fully be addressed by standardization and disclosure is that of extending the geographic reach of issuances and green-bonds-eligible projects to emerging and developing countries. In developing countries, weak capital markets and low credit ratings for potential green bonds issuers limit the geographic reach of green bonds markets – and green bonds' contribution to sustainable development.

In fact, green bonds proceeds have rarely been allocated to low-income countries (LICs). Development banks like the IFC have focused on middle-income countries. There are several reasons that green bonds are rarely allocated to LICs, for instance, the institutions responsible for LICs at some of the major multilateral DBs, such as the World Bank's International Development Association and the AfDB's African Development Fund, do not issue bonds in capital markets because they use aid allocations to finance projects. Another reason is that in poorer countries, projects that could be eligible for financing through green bonds are often just too small.

However, as we have noted, green bonds have been issued in several emerging economies and the geographic reach of green bonds markets is expanding. As green bonds markets develop in emerging economies, enhanced standardization and disclosure rules will be as important for them as for the global green bonds market.¹¹ In fact, in developing and emerging countries, where capital markets are weakly developed and governance structures are just now being designed, there may be a window of opportunity for developing green standards since there are no costs from changing governance structures and there may also be fewer vested interests that benefit from weak governance.

4. CONCLUSION AND RECOMMENDATIONS: QUO VADIS, GREEN BONDS MARKET?

In the past decade there has been an explosive growth in the market for green bonds. Government efforts to tackle global warming, the increase in the number of socially responsible investors, rising concerns about stranded assets and great enthusiasm in debt capital markets because of extremely low interest rates have all contributed to the popularity of green bonds as an asset class. Recognition that environmental and financial issues are intimately linked has also grown.

To realize the enormous potential of green bonds, several challenges must be addressed, notably the risk of greenwashing, limited standardization, an undiversified investor base, the time lag from directing proceeds

from a bond sale to green projects or assets, narrow secondary markets, the high costs of green bonds issuance and the development of a green project pipeline. An improved governance framework could help to address several of these challenges. What should policymakers, issuers, investors and other groups who seek to contribute to the greening of bond markets do? We see considerable potential for reforms in six areas.

4.1 Transparency and Disclosure

The first area is transparency and disclosure. For the green bonds market to contribute to ecologically sustainable development, there must be greater transparency about how green bonds are defined. For instance, the GBPs should be elaborated to provide a clearer definition of green bonds that leaves less room for greenwashing. Second opinions could provide more detailed assessments of the greenness of particular projects the way Cicero does with its ‘three shades of green’ approach (Clapp et al., 2016; Nanji et al., 2014; Schneeweiß, 2016). Cicero’s approach should be emulated. To make it easier to compare second opinions, the evaluation criteria – including information about the issuer’s core business activities – and their rating scales must be standardized and adopted by all second opinion providers.

Disclosure reforms should aim to ensure publication of details about the intended use of proceeds and the annual reporting of the actual use of proceeds and second opinions. Covered bonds could increase transparency because they are more strictly regulated than uncovered bonds.

Green bonds indices may also increase transparency on the green bonds market for both issuers and investors by defining specific criteria.¹² Indeed, more green bonds indices have been established, such as those of S&P Dow Jones, Solactive, Barclays MSCI and Bank of America Merrill Lynch. One good example is the Barclays MSCI Green Bond Index which is based on various criteria for green bonds, including the definition of the eligible use of proceeds and the evaluation and project selection methods; it also stipulates regular reporting (Reichelt and Davies, 2015). Other examples are the S&P Green Project Bond and the S&P Green Bond indices (Standard and Poors, 2016). That said, more transparency and clarity is needed with regard to the selection criteria used to compile indices.

4.2 Compliance with Standards

The second area concerns compliance with standards. Several measures could help to improve that. For instance, bodies that design green bonds

indices could more closely monitor the bonds included in any index of bonds that are claimed to adhere to the strictest industry standards. Rating agencies could explicitly consider the environmental aspects of green bonds in their evaluations. Financial authorities could regulate the green bonds market, and institutional investors could integrate green investment criteria in their guidelines.

Industry self-regulators should stop recommending compliance with standards and require it. Public disclosure will help to ensure compliance. If the green bonds market is to mature with integrity, clearer monitoring responsibilities must be defined, with sanctions for violating the rules. It will be equally important to eliminate market monitors' conflicts of interest. For instance, rating agencies and some second opinion providers could face a conflict of interest because of their need to maintain good relations with the bond issuers who commission the opinions.

4.3 Reduction of Complexity

The third area regards the need to reduce the complexity of the green bonds asset class. Financial products must be simple to attract investors. Green bonds' growing diversity, which results from more and more issuers in a broad range of sectors designing their own bonds, must somehow be reversed in order to make green bonds appeal to the mass market.

4.4 Cost of Green Bonds Issuance

The fourth area relates to the costs of issuing green bonds. Smaller green projects may be aggregated to enable lower-cost access to the green bonds market. Tax benefits for green bonds issuers or investors are one way to reduce costs. However, cost reductions for green bonds issuance should be granted to small and medium-sized companies rather than to large entities.

According to a senior banker at HSBC, larger institutions with more sophisticated funding operations, such as banks, can more easily incorporate green bonds into their borrowing plans (Jackson, 2016). The informant argues that larger institutions can spread the cost of running a green policy over many issues by maintaining a presence in the bond markets. Public schemes to reduce the price of green bonds must also be weighed against other climate policies (Shishlov et al., 2016).

4.5 Research

The fifth important area is the support of research. There is very little academic research which links developments in green bond markets to wider

theories of finance. Few scholars have examined the political and economic drivers of green bonds market development and the political and economic consequences of green bonds issuances. Among the important research questions are: What are the risks of aggregating green bonds projects through securitization? To what extent does the green bonds market help to mobilize additional climate finance? Which governance mechanisms would help the green bonds market to develop qualitatively? Scholars from business and management studies, economics and international politics could cluster around these questions. It will also be important to track green bonds financing provided by and in developing economies – ‘South-South climate finance’ – in order to foster such flows and more effectively align them with traditional climate finance (Ha et al., 2016).

4.6 Expand Governance Inclusiveness

Sharing more detailed information between the various stakeholders will make governance more inclusive. Round tables of investors, issuers, governments, NGOs and regulators should be organized to discuss what and how much regulation and policy support is needed to foster the development of green bonds. Such an approach could help the market to mobilize climate finance while also meeting the needs of the finance industry. The establishment of the GBP Executive Committee was an important step in the right direction. The committee gathers together issuers, investors and intermediaries in the green bonds market to assess its development and formulate best practices.

The ICMA similarly operates as a secretariat supporting the GBP, for example, by facilitating the exchange of information among stakeholders and offering advice about its governance (ICMA, 2017). However, it will be important to ensure that stakeholder dialogues are inclusive. One recent initiative is the CBI Climate Bond Standards, where working groups consisting of technical and industry experts and an advisory board of institutional investors and environmental NGOs collaborate to develop standards. When the development of standards becomes a more inclusive process, the resulting standards may also be considered more legitimate.

NOTES

1. Note that green bonds do not necessarily imply a limited exposure of investors to environmentally related financial risks such as Ehlers and Packer (2016) highlight.
2. There are notable exceptions. See for instance Bracking (2015), duPont et al. (2015) and Flaherty et al. (2016).
3. For instance, both the World Bank and its green use-of-proceeds bonds are rated ‘AAA’.

4. The volume of the worldwide bond market amounts to about USD 100 trillion (Schneeweiß, 2016).
5. See also Dietz et al. (2016) for the impact of climate change on asset values.
6. Interview with a senior official from the African Development Bank, Abidjan, 21 January 2016. Interview with a senior official of the KfW in Bonn, 13 November 2015.
7. Interview with a senior official of the KfW in Bonn, 13 November 2015.
8. Interviews with senior officials from the KfW, the EIB and the AfDB.
9. See for instance Ceres (2015), Cort and Krosinsky (2015), KPMG (2015) and OECD (2015). The Paris Green Bonds Statement, a document published by existing and potential green bonds investors, also emphasizes that enhanced standardization and disclosure rules are essential for the green bonds market to develop (CBI, 2015b).
10. There may be exceptions. For instance, the AfDB reports that feedback from investors indicates that AfDB's Swedish Krona green bonds issuance was executed at a price that would not have been possible without a green label (Duru and Nyong, 2016: 4).
11. This holds notably for green bonds targeted at financing projects which support climate change adaptation, which is of major concern to developing and emerging economies. Because the policy debate and bond issuance have focused on green credentials in relation to climate-change mitigation, the criteria for assessing the climate-change adaptation benefits of a green bond have received little attention in international discussions about standardization. For instance, as of June 2015, the IBRD had committed 78 per cent of green bonds proceeds to mitigation and 22 per cent to adaptation projects (World Bank, 2015a).
12. Extant research suggests that responsible investment indices like the FTSE4Good Index might also have a positive impact on environmental management (Mackenzie et al., 2013).

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